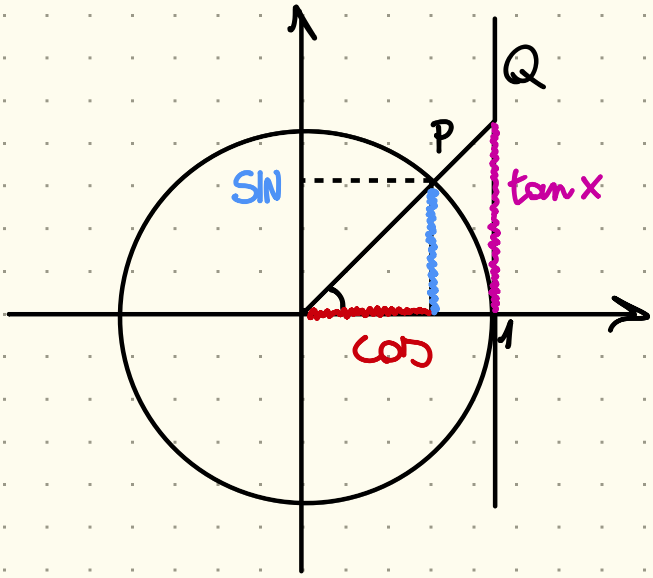
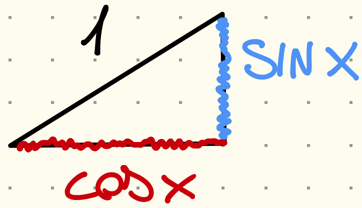
IDENTITÀ TRIGONOMETRICHE

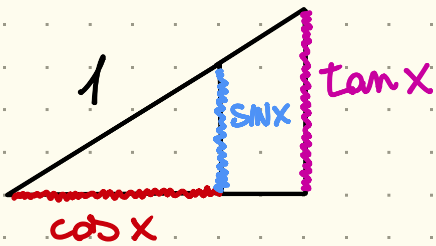




$$\Downarrow$$

$$\cos^2 x + \sin^2 x = 1$$

$$(\cos x)^2 \neq \cos(x^2)$$



$$\Rightarrow \frac{\tan x}{1} = \frac{\sin x}{\cos x}$$

$$\Rightarrow \tan x = \frac{\sin x}{\cos x}$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

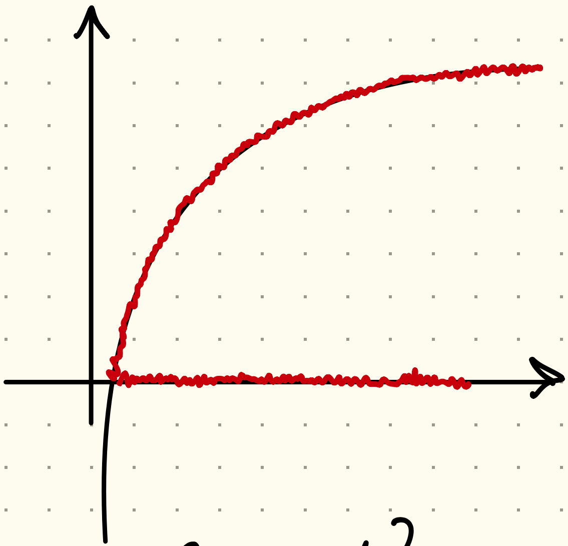
105) **Qual è il dominio della funzione** $f(x) = \sqrt{\ln(2 - 2\cos^2 x)}$**?**

• $\text{Dom } f = ?$

• $2 - 2\cos^2 x > 0$

• $\log(\dots) \geq 0$

$\iff 2 - 2\cos^2 x \geq 1$



• $\text{Dom } f = \{x \in \mathbb{R} : 2 - 2\cos^2 x \geq 1\}$

• $2 - 2\cos^2 x \geq 1$

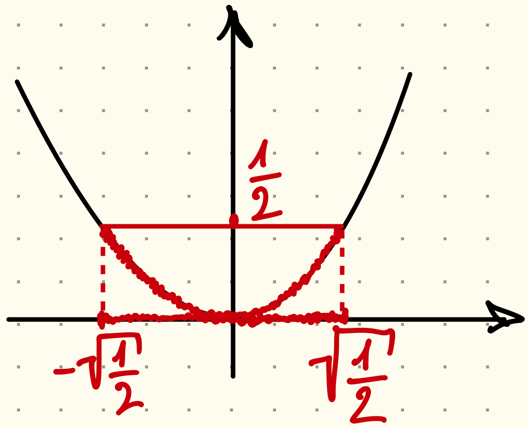
$2 - 1 \geq 2\cos^2 x$

$2\cos^2 x \leq 1$

$\cos^2 x \leq \frac{1}{2}$

$\Downarrow$

$\frac{1}{2} \leq \cos x \leq \sqrt{\frac{1}{2}}$



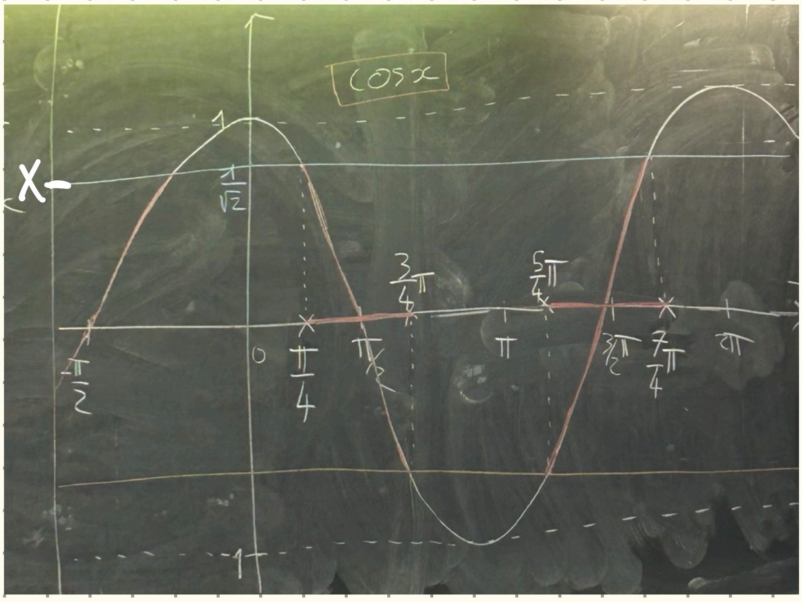
$\Downarrow$

$-\frac{1}{\sqrt{2}} \leq \cos x \leq \frac{1}{\sqrt{2}}$

• $\cos^2 x = \frac{1}{2}$

$$\underbrace{\cos^2 x}_{\frac{1}{2}} + \underbrace{\sin^2 x}_{\frac{1}{2}} = 1$$

$$\frac{1}{2} + \frac{1}{2} = 1$$



• In $[0, 2\pi]$

$$\left[\frac{\pi}{4}, \frac{3}{4}\pi\right] \cup \left[\frac{5}{4}\pi, \frac{7}{4}\pi\right]$$

OSS) $\cos^2 x + \sin^2 x = X \quad \forall x \in \mathbb{R}$

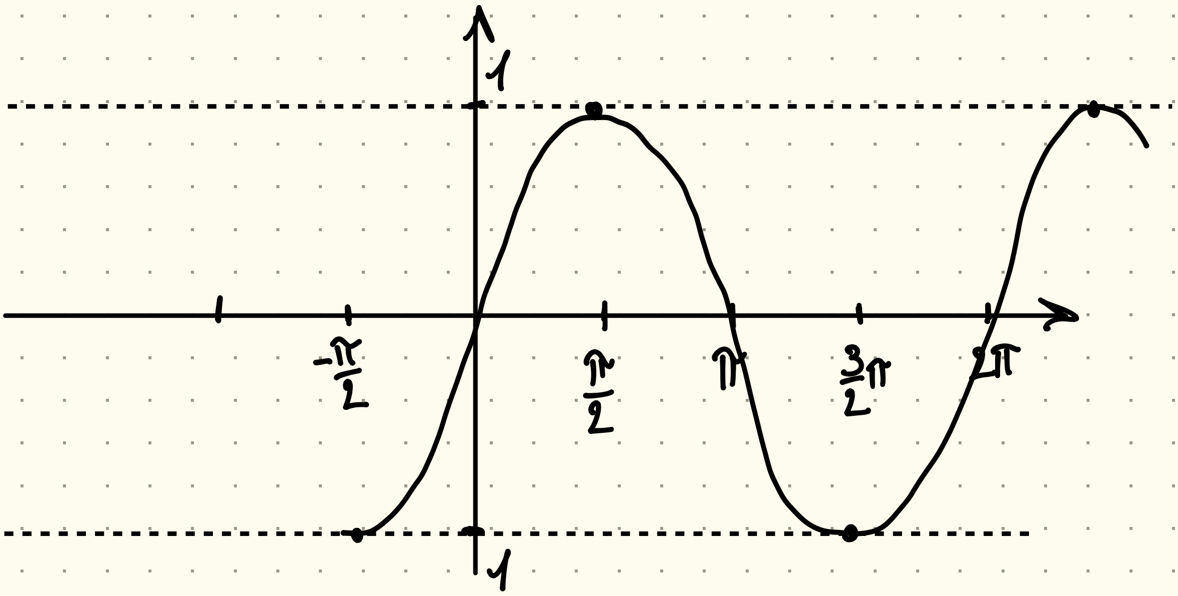
$\cos^2 x = 1 - \sin^2 x \quad \forall x \in \mathbb{R}$

$\cos x = \sqrt{1 - \sin^2 x} \quad \cancel{\forall x \in \mathbb{R}} \quad \forall x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\boxed{\sqrt{y^2} \neq y}$

FUNZIONI TRIGONOMETRICHE INVERSE

- SIN NON INIETTIVA, QUINDI NON INVERTIBILE



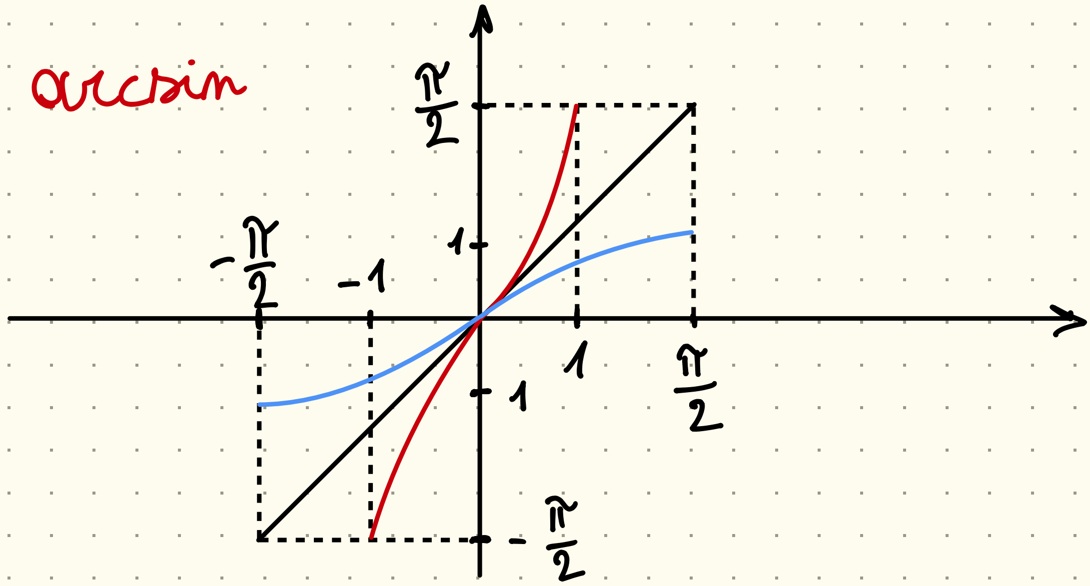
- PER RENDERLA INVERTIBILE RESTRINGIAMO IL SENO a $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ e

$\sin\big|_{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]}$ $\Rightarrow$ É INIETTIVO E INVERTIBILE:

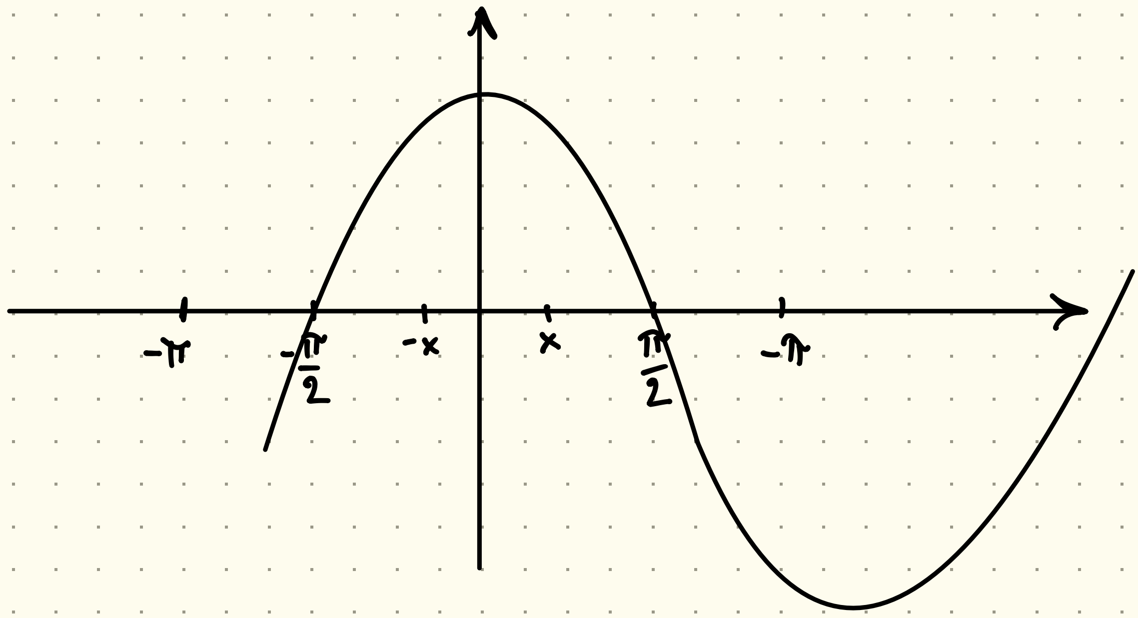
- $\arcsin := \left(\sin\big|_{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]}\right)^{-1} =$
- $\text{Dom} \arcsin = \text{Im}\left(\sin\big|_{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]}\right) = [-1, 1]$
- $\text{Im} \arcsin = \text{Dom}\left(\sin\big|_{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]}\right) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

• GRAFICO ARCSIN

arcsin



• COSENO



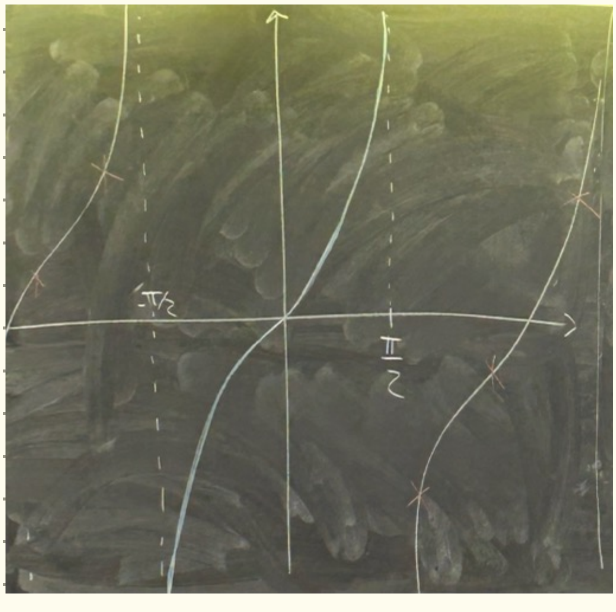
• COS NON É INIETTIVA, NON INVERTIBILE

• $\cos\big|_{[0,\pi]}$ É INIETTIVA E QUINDI INVERTIBILE

- DEFINIAMO
  - $\arccos \left(\cos|_{[0,\pi]} \right)^{-1}$
  - $\text{Dom} \arccos = \text{Im}\left(\cos|_{[0,\pi]} \right) = [-1,1]$
  - $\text{Im} \arccos = \text{Dom}\left(\cos|_{[0,\pi]} \right) = [0,\pi]$

DISEGNO

- TANGENTE
  - tan NON É INIETTIVA, QUINDI NON INVERTIBILE
  - $\tan|_{\left(-\frac{\pi}{2},\frac{\pi}{2}\right)}$ => É INIETTIVA



- $\arctan = \left(\tan|_{(-\frac{\pi}{2}, \frac{\pi}{2})} \right)^{-1}$

- $\operatorname{Dom} \arctan = \operatorname{Im} (\;) = \mathbb{R}$

- $\operatorname{Im} \arctan = \operatorname{Dom} (\;) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$